

Medical Assistive Robot

Submitted By-

Meher Afroz

ID: 22201176

Email: 22201176@uap-bd.edu

Hridita Lasker

ID: 22201184

Email: 22201184@uap-bd.edu

Jannatul Maowa Meghla

ID: 22201194

Email: 22201194@uap-bd.edu

Submitted To-

Dr. Nazmun Nahid

Assistant Professor



**Department of Computer Science & Engineering
University of Asia Pacific**

June 2026

Acknowledgements

All praise and gratitude belong to Almighty Allah, the Most Gracious and the Most Merciful, for bestowing upon us the knowledge, patience, and determination required to undertake and complete this work.

We consider it a privilege to express our sincere and heartfelt appreciation to all those who contributed, directly or indirectly, to the successful completion of this project.

We would like to place on record our deepest sense of gratitude to our respected supervisor, Dr. Nazmun Nahid, Assistant Professor, Department of Computer Science and Engineering, University of Asia Pacific, for her invaluable guidance, scholarly insight, and unwavering support throughout the entire course of this project. Her constructive criticism, thoughtful suggestions, and constant encouragement were the driving forces behind every milestone we achieved. We are truly fortunate to have had the opportunity to work under her supervision.

We wish to acknowledge the dedication and collaborative spirit of each of our team members, whose consistent effort, mutual cooperation, and shared commitment made the realization of this project possible.

Abstract

This paper presents Medical Assistive Robot, an intelligent mobile robotic system designed to support healthcare workers in medical waste management. The system uses ultrasonic and IR sensors to follow nurses or hospital staff automatically and maintain a safe distance during movement. It also includes an automatic lid-opening mechanism that allows touchless medical waste disposal when a hand or waste is detected near the bin. The robot is controlled by a microcontroller that processes sensor data and controls motors, direction, and servo-based lid operation. Experimental validation confirms successful integration of sensing, movement, and automatic disposal functions, demonstrating a simple, hygienic, and practical solution for healthcare environments.

Index Terms - Medical Assistive Robot, Human-Following Robot, Smart Waste Management, Ultrasonic Sensor, IR Sensor, Automatic Lid, Microcontroller, Healthcare Robotics.

Contents

List of Figures	viii
List of Tables	ix
1 Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement	1
1.3 Objectives of the Project	2
1.4 Scope of the Work	2
1.5 Expected Outcomes	2
1.6 Impacts of the Project	3
1.6.1 Impact on Societal and Cultural Issues	3
1.6.2 Impact on Health, Safety, and Legal Issues	4
1.6.3 Impact on Environment and Sustainability Issues	4
1.7 Significance	4
2 Background and Literature Review	5
2.1 Preliminaries	5

2.2	Related Works / Existing Systems	6
2.3	Comparative Analysis of Related Works	7
2.4	Research Gap / Limitations of Existing Methods	8
2.5	Summary	8
3	System Analysis and Design	9
3.1	Requirement Analysis	9
3.1.1	Functional Requirements	9
3.1.2	Non-Functional Requirements	10
3.2	System Overview / Architecture	11
3.2.1	High-Level Architecture	12
3.2.2	Embedded Diagram	13
3.3	Hardware Components	14
3.3.1	Functionalities of the Components	15
3.3.2	Circuit Design	16
3.4	Robot Structure Details	17
3.5	Design Alternatives and Rationale	18
3.6	Summary	18
4	Methodology and Implementation	19
4.1	Methodological Framework	19
4.2	Tools, Libraries, and Technologies Used	20

4.3	Flowchart and Algorithm Development	22
4.3.1	Flow Chart	23
4.3.2	Algorithm Development	23
4.4	Code	24
4.4.1	Main Modules	30
4.5	Summary	32
5	Results and Evaluation	33
5.1	Working Principle	33
5.1.1	Data Flow Diagram	34
5.2	Planning and Testing	35
5.2.1	Module Classification and Planning	35
5.2.2	Sensor and Actuator Testing	35
5.3	Assembly and Integration	35
5.4	Prototype	36
5.5	Final Design	37
5.5.1	Additional Features and Purposeful Engineering Decisions . .	37
5.6	Performance Evaluation	38
5.7	Result Analysis	38
5.8	Summary	40
6	Project Planning and Budget	41

6.1	Project Timeline	41
6.1.1	Gantt Chart	42
6.2	Budget Estimation	42
6.3	Summary	44
7	Standards and Design Constraints	45
7.1	Compliance with Standards and Professional Practice	45
7.1.1	Software Standards	45
7.1.2	Hardware Standards	45
7.1.3	Safety and Ethical Considerations	46
7.1.4	Environmental and Sustainability Considerations	46
7.1.5	Professional Engineering Practice	46
7.2	Design Constraints	46
7.2.1	Technical Constraints	47
7.2.2	Economic Constraints	47
7.2.3	Operational Constraints	47
7.2.4	Health and Safety Constraints	47
7.3	Complex Engineering Problem (CEP) Coverage	47
7.4	Complex Engineering Activities (CEA) Coverage	49
7.5	Knowledge Profile Mapping	50
7.6	Summary	50

8	Conclusion and Future Work	52
8.1	Conclusion	52
8.2	Future Work	53
	Appendices	55
A	Surveys	55
A.1	Survey 1	55
A.2	Survey 2	56
B	Team Contributions	59
C	Ethics and Compliance Statements	60

List of Figures

3.1	Work-flow of the Medical Assistive Robot	12
3.2	Embedded Diagram of the Medical Assistive Robot	13
3.3	Hardware Components Used in the Medical Assistive Robot	14
3.4	Circuit Design of Human following car of the Medical Assistive Robot	16
3.5	Circuit Design of Automatic Dustbin of the Medical Assistive Robot .	16
4.1	Flowchart of the Medical Assistive Robot	23
5.1	Working principle of the Medical Assistive Robot	34
5.2	Data Flow Diagram of the Medical Assistive Robot	34
5.3	Prototype of the Medical Assistive Robot	36
6.1	GanttChart of the Medical Assistive Robot	42
8.1	Prototype feedback interview with Nurse	58

List of Tables

2.1	Comparative Analysis of Related Works	7
4.1	Tools, Libraries, and Technologies Used	21
5.1	Functional Test Result	39
6.1	Project Timeline	42
6.2	Estimated Project Budget	43
7.1	Mapping with Complex Engineering Problems (Ps)	48
7.2	Mapping with Complex Engineering Activities (As)	49
7.3	Knowledge Profile Mapping	50
7.4	Overall Mapping of Knowledge Profile, CEP and CEA	51
8.1	Future Work of the Medical Assistive Robot	53
8.2	Team Contributions in Medical Assistive Robot Project	59

Chapter 1

Introduction

1.1 Background and Motivation

Medical waste handling in hospitals and clinical environments often requires direct human involvement. This may increase hygiene risks, physical workload, and the possibility of cross-contamination. Healthcare workers frequently move between patients and waste disposal points, which can make the process time-consuming and less efficient. Recent developments in robotics, sensors, and microcontroller-based automation have enabled the development of smart assistive systems for safer and more hygienic medical waste management.

1.2 Problem Statement

Existing medical waste bins are usually static and require healthcare workers to move toward them for waste disposal. In busy hospital environments, this can reduce efficiency and increase the risk of physical contact with contaminated surfaces. A smart assistive robot that can follow healthcare workers and provide touchless waste disposal is needed to improve hygiene, reduce manual effort, and support safer waste management.

1.3 Objectives of the Project

- Design a human-following robot for healthcare environments.
- Use ultrasonic and IR sensors to detect user distance and direction.
- Implement automatic lid control for touchless medical waste disposal.
- Control robot movement using a microcontroller-based system.
- Reduce physical effort and minimize cross-contamination risks.

1.4 Scope of the Work

The current prototype has several limitations that can be addressed in future versions to improve performance, accuracy, and real-world usability.

- Better obstacle avoidance: Additional sensors will be added on all sides to detect and avoid unexpected obstacles in busy hospital corridors.
- IoT monitoring: An ESP32 module will enable wireless data transmission to a cloud dashboard, providing real-time status updates and alerts for hospital staff.
- Camera-based person-following: A camera module with computer vision will replace IR-based tracking for more accurate and reliable identification of the target healthcare worker.
- Stronger chassis and longer battery: The robot frame will be upgraded to support heavier loads, and a higher-capacity battery will extend runtime to cover a full hospital shift.

1.5 Expected Outcomes

The expected outcome of this project is to develop a Medical Assistive Robot that can effectively support healthcare workers in their daily activities. The robot is expected

to follow a healthcare worker while maintaining a safe distance and moving in the correct direction based on sensor inputs.

Another expected outcome is the implementation of a touchless medical waste disposal system. The automatic lid mechanism is expected to reduce direct contact with waste bins, helping to improve hygiene and minimize the risk of infection.

The project is also expected to demonstrate the practical use of sensors, motors, and microcontrollers in healthcare-related robotic applications. Through successful operation, the robot should reduce manual effort, improve convenience, and support safer waste management practices.

Overall, the project aims to showcase a low-cost and efficient robotic solution that can contribute to smarter and more technology-driven healthcare environments.

1.6 Impacts of the Project

The Medical Assistive Robot is designed to support healthcare workers by carrying medical supplies and providing touchless waste disposal. It helps reduce manual effort, improves hygiene, and promotes the use of smart technologies in healthcare environments. The project demonstrates how robotics can be used to make healthcare services safer and more efficient.

1.6.1 Impact on Societal and Cultural Issues

This project can contribute to the modernization of healthcare services by introducing robotic assistance in hospitals and clinics. It helps reduce the workload of healthcare workers, allowing them to focus more on patient care. The robot also encourages the adoption of innovative technologies and increases public awareness about the benefits of automation in healthcare. As a result, it supports the development of a more technology-driven and efficient society.

1.6.2 Impact on Health, Safety, and Legal Issues

The Medical Assistive Robot promotes better hygiene by providing a touchless waste disposal system, which helps reduce the risk of spreading infections. By carrying medical items and following healthcare workers, it can reduce physical strain and improve workplace safety. Since the robot does not collect personal patient information, it minimizes privacy concerns. However, proper testing and safety measures should be followed before using similar systems in real healthcare environments.

1.6.3 Impact on Environment and Sustainability Issues

The robot can support proper medical waste handling through its dedicated disposal mechanism, helping maintain a cleaner healthcare environment. By encouraging organized waste management, it may contribute to reducing environmental contamination. The use of low-power electronic components also supports energy-efficient operation. In the future, rechargeable batteries and eco-friendly materials could further improve the sustainability of the system.

1.7 Significance

The Medical Assistive Robot can be useful in hospitals, clinics, and healthcare facilities where hygienic waste disposal is important. It helps reduce direct contact with waste bins and supports healthcare workers by following them during their tasks. The project also demonstrates the practical use of embedded systems, sensors, motor control, and robotics in solving real-life healthcare problems.

Chapter 2

Background and Literature Review

This chapter presents the theoretical background and related studies relevant to the Medical Assistive Robot. It discusses the basic concepts of human-following robots, medical waste management, and touchless disposal systems. The chapter also reviews existing research works and identifies the research gap that motivates the development of the proposed system.

2.1 Preliminaries

Preliminaries A Medical Assistive Robot is a robotic system designed to support healthcare workers in performing routine tasks more efficiently. Human-following robots use sensors to detect and track a person's movement, allowing the robot to move along with the user automatically.

In healthcare environments, proper medical waste handling is important for maintaining hygiene and preventing contamination. Touchless waste disposal systems help reduce direct contact with waste bins and improve safety. By combining human-following technology with automatic lid control, a medical assistive robot can support healthcare workers and improve waste management practices.

2.2 Related Works / Existing Systems

Human-following robots have been extensively studied for service and healthcare applications. A comprehensive review presented in [1] discussed various human-following techniques, including vision-based, sensor-based, and hybrid approaches. The study highlighted the advantages and limitations of different tracking methods and emphasized their applications in assistive robotics.

A sensor-based human-following robot was proposed in [2], where distance and direction information were used to track and follow a target user. The system demonstrated reliable human-following performance in indoor environments and provided a foundation for low-cost autonomous robotic platforms.

In the field of waste management, the authors in [3] developed a smart waste management robot capable of automated waste collection and disposal. Their work demonstrated the effectiveness of sensor-based automation in reducing human involvement and improving operational efficiency.

A recent study by Zhou et al. [4] introduced a human-following system that operates without a prior map of the environment. The proposed method combined multiple sensing technologies to achieve robust and accurate tracking in dynamic environments. The research showed significant improvements in navigation and human-following performance.

Although these studies have addressed human-following and automated waste management separately, limited work has integrated both functionalities into a single healthcare-oriented robotic platform. Therefore, the proposed Medical Assistive Robot combines human-following capability with automatic waste disposal to provide a practical solution for hospital and healthcare environments.

2.3 Comparative Analysis of Related Works

Table 2.1: Comparative Analysis of Related Works

Ref.	Authors / Study	Main Technology	Human Following	Waste Management	Special Feature	Relevance to Proposed Project
[1]	Islam et al. (2018)	Vision-based, sensor-based and hybrid tracking	Yes	No	Comprehensive review of human-following techniques	Provides theoretical foundation for human-following robots.
[2]	Wang et al. (2017)	Multi-sensor human tracking	Yes	No	Accurate real-time human tracking	Similar objective of following a user using sensor inputs.
[3]	Bonifacio (2025)	Medical waste management robotics	No	Yes	Automated waste collection and handling	Directly related to medical waste disposal functionality.
[4]	Zhou et al. (2024)	Multi-sensor navigation and tracking	Yes	No	Human following in unknown environments	Useful for autonomous navigation and tracking concepts.

The reviewed studies mainly focus on either human-following robots or medical waste management systems. Islam et al. [1] provided a broad overview of different human-following techniques and their applications in assistive robotics. Wang et al. [2] demonstrated accurate real-time human tracking using multiple sensors, which is closely related to the human-following functionality of the proposed system.

Bonifacio [3] explored the use of robotic systems in medical waste collection, classification, and treatment. This work highlights the importance of automation in improving hygiene and reducing human involvement in waste handling. However, the system did not include human-following capability.

Zhou et al. [4] developed a human-following system capable of operating without a prior map of the environment. The study showed improved tracking and navigation performance in dynamic environments, but it did not address medical waste management.

From the comparison, it can be observed that existing studies generally focus on a single functionality. Some focus on human-following behavior, while others focus on waste management automation. The proposed Medical Assistive Robot combines

both features into a single healthcare-oriented platform. It can follow healthcare workers and provide touchless medical waste disposal, making it a practical and low-cost solution for hospitals and clinical environments.

2.4 Research Gap / Limitations of Existing Methods

Although many human-following robots and smart waste management systems have been developed, several limitations still exist. Many solutions require expensive sensors, complex image processing, or advanced computing hardware. Some systems are difficult to implement in small-scale healthcare environments due to their cost and complexity.

There is a need for a simple, affordable, and efficient robotic system that can follow healthcare workers while also providing touchless medical waste disposal. The proposed Medical Assistive Robot addresses this gap by combining human-following movement and automatic lid control using low-cost sensors, motors, and a microcontroller-based design.

2.5 Summary

This chapter discussed the background concepts and previous research related to human-following robots and medical waste management systems. The literature review showed that existing solutions provide useful features but often focus on only one function or require costly technologies. The identified research gap highlights the need for a low-cost and practical Medical Assistive Robot that combines human-following and touchless waste disposal features. The next chapter presents the system analysis and design of the proposed robot.

Chapter 3

System Analysis and Design

3.1 Requirement Analysis

Requirement analysis is an important step in designing and developing the Medical Assistive Robot. It helps identify the necessary features and system requirements needed to achieve the project objectives. The robot is designed to assist healthcare workers by following them and providing touchless medical waste disposal. Therefore, the system requires sensors for detecting movement and distance, motors for mobility, a servo motor for automatic lid control, and a microcontroller to coordinate all operations. The identified requirements ensure that the robot can operate efficiently, safely, and reliably in healthcare environments.

3.1.1 Functional Requirements

The functional requirements describe the specific tasks that the Medical Assistive Robot must perform.

- The robot shall detect and follow a healthcare worker using sensor inputs.
- The robot shall maintain a safe distance while following the user.
- The robot shall identify changes in the user's direction and adjust its movement accordingly.
- The robot shall move forward, left, or right based on sensor readings.

- The robot shall stop when the user is not detected or when an obstacle is present.
- The robot shall automatically open the waste bin lid when waste is brought near the disposal area.
- The robot shall automatically close the lid after a short period.
- The robot shall continuously monitor sensor data and update its actions in real time.
- The robot shall support safe and hygienic medical waste disposal.

3.1.2 Non-Functional Requirements

The non-functional requirements describe the quality attributes and operational constraints of the system.

- The system should be low-cost and suitable for educational and prototype purposes.
- The robot should operate reliably under normal indoor healthcare conditions.
- The system should be easy to use and require minimal user interaction.
- The robot should respond quickly to sensor inputs and movement changes.
- The design should be compact and portable for easy movement.
- The system should consume low power during operation.
- The robot should maintain stable performance during continuous operation.
- The system should be easy to maintain and troubleshoot.
- The design should prioritize user safety and hygienic operation.
- The robot should be scalable for future improvements and additional features.

3.2 System Overview / Architecture

The Medical Assistive Robot is designed to assist healthcare workers by combining human-following functionality and touchless medical waste disposal within a single robotic platform. The system consists of sensing, processing, mobility, and waste management units that work together to perform the required tasks efficiently.

At the core of the system is a microcontroller, which serves as the central control unit. It continuously receives data from the ultrasonic sensor and IR sensors, processes the information, and makes decisions based on the programmed control logic. The sensors are responsible for detecting the position, movement, and direction of the healthcare worker, allowing the robot to follow the user while maintaining a safe distance.

The mobility unit consists of DC motors and a motor driver. Based on the sensor data, the microcontroller controls the motors to move the robot forward, turn left, turn right, or stop when necessary. This enables the robot to navigate smoothly and respond to changes in the user's movement.

The waste management unit includes an ultrasonic sensor and a servo motor connected to the waste bin lid. When an object or waste item is detected near the disposal area, the microcontroller activates the servo motor to automatically open the lid. After a short period, the lid closes automatically, reducing physical contact and improving hygiene.

The system is powered by a portable power supply, allowing the robot to operate independently. All hardware components are integrated to ensure reliable communication and coordinated operation. The overall architecture is designed to be simple, cost-effective, and suitable for healthcare environments where assistance, safety, and cleanliness are important.

By combining human-following capability with automated waste disposal, the Medical Assistive Robot provides a practical solution that supports healthcare workers, reduces manual effort, and promotes better hygiene practices in hospitals and clinical settings.

3.2.1 High-Level Architecture

This diagram provides a high-level view of the Medical Assistive Robot system. It illustrates how data is collected from the ultrasonic and IR sensors and processed by the microcontroller to control the robot's movement. The diagram also shows how the robot follows a healthcare worker while maintaining a safe distance and how the automatic waste disposal mechanism operates. Through the coordination of sensors, motors, and control logic, the system supports both human-following functionality and touchless medical waste disposal in a healthcare environment.

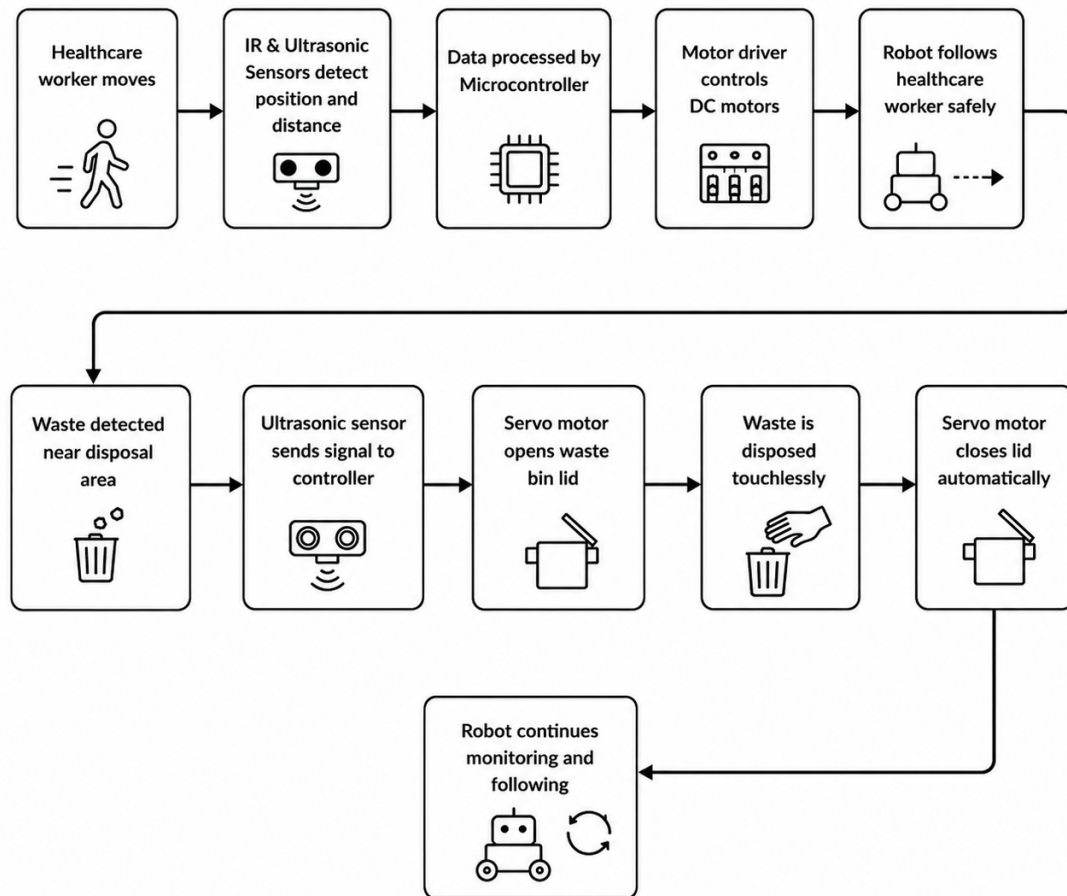


Figure 3.1: Work-flow of the Medical Assistive Robot

3.2.2 Embedded Diagram

The embedded diagram presents the overall hardware structure of the proposed Medical Assistive Robot and illustrates how the different components work together. The microcontroller serves as the central control unit of the system, receiving data from the ultrasonic and IR sensors and processing it according to the programmed logic.

The sensors are responsible for detecting the position and movement of the healthcare worker, enabling the robot to follow the user while maintaining a safe distance. Based on the processed sensor data, the microcontroller controls the motor driver and DC motors to perform the required movements.

The robot also includes an automatic waste disposal mechanism. When an object is detected near the waste bin, the microcontroller activates the servo motor to open the lid automatically. After a short period, the lid closes automatically to maintain hygiene and reduce physical contact.

All components are powered through a common power supply and operate together to provide human-following and touchless waste disposal functionalities. The integrated design ensures efficient, reliable, and low-cost operation suitable for healthcare environments.

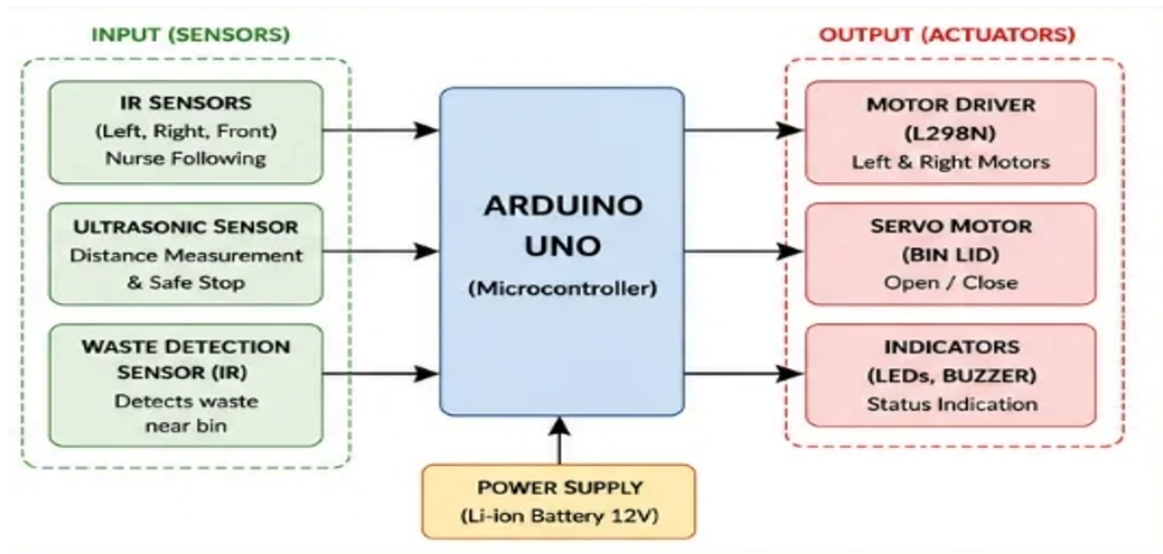


Figure 3.2: Embedded Diagram of the Medical Assistive Robot

3.3 Hardware Components



(a) Arduino Uno



(b) HC-SR04



(c) IR Sensor



(d) Chassis



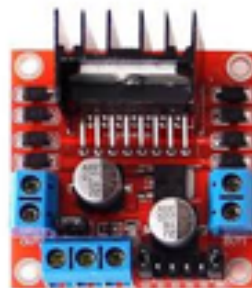
(e) Servo Motor



(f) Battery



(g) Breadboard



(h) L298N Motor Driver



(i) DC Motor



(j) Breadboard

Figure 3.3: Hardware Components Used in the Medical Assistive Robot

Figure 3.3 shows the hardware components used in the Medical Assistive Robot. These components work together to enable the robot's human-following and touchless waste disposal functionalities. The Arduino Uno acts as the main controller, while the ultrasonic and IR sensors are used for distance and movement detection. The DC motors, motor driver, and chassis provide mobility to the robot. A servo motor is used to control the automatic opening and closing of the waste bin lid. The breadboard and battery support circuit connections and power supply, ensuring the proper operation of the entire system.

3.3.1 Functionalities of the Components

- Microcontroller: Works as the main control unit of the robot.
- Ultrasonic Sensor: Measures the distance between the robot and the user.
- IR Sensors: Detect left and right direction for human-following movement.
- DC Motors: Allow the robot to move forward and turn.
- Motor Driver: Controls motor speed and direction.
- Servo Motor: Opens and closes the dustbin lid.
- Battery: Provides power to the whole system.
- Chassis and Wheels: Support movement and hold all hardware components.
- Dustbin Body and Lid: Used for medical waste collection and touchless disposal.
- Wires and Connectors: Connect all electronic components properly.

3.3.2 Circuit Design

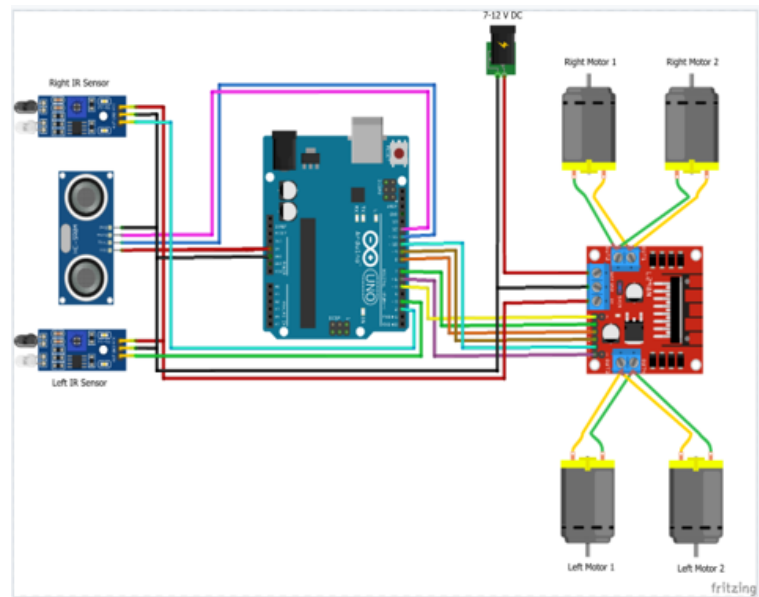


Figure 3.4: Circuit Design of Human following car of the Medical Assistive Robot

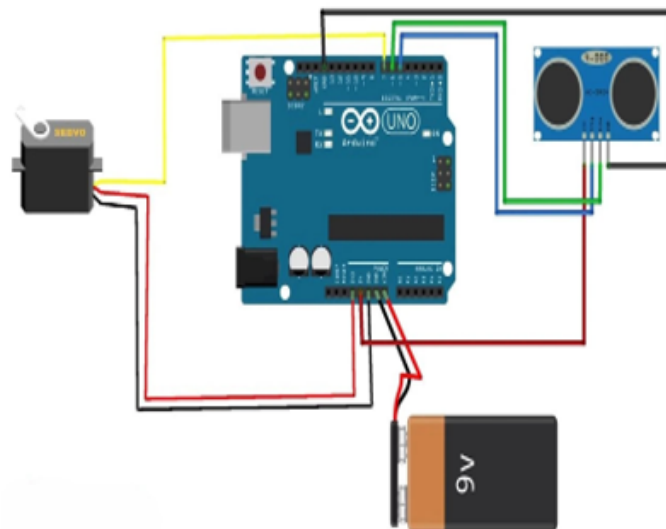


Figure 3.5: Circuit Design of Automatic Dustbin of the Medical Assistive Robot

Circuit Design Description

The circuit design of the Medical Assistive Robot consists of two main subsystems: the human-following module and the automatic waste disposal module. The human-following module uses an Arduino Uno, ultrasonic sensor, IR sensors, L298N motor driver, and DC motors to detect the position of the healthcare worker and control the movement of the robot accordingly. The automatic waste disposal module consists of an Arduino Uno, ultrasonic sensor, servo motor, and power supply. When an object is detected near the waste bin, the ultrasonic sensor sends a signal to the controller, which activates the servo motor to open the lid automatically. These circuits are integrated to enable efficient human-following behavior and touchless medical waste disposal within a single robotic platform.

3.4 Robot Structure Details

The Medical Assistive Robot is designed with a compact and lightweight structure suitable for indoor healthcare environments. The robot is built on a four-wheel chassis that provides stability and smooth movement during operation. The Arduino Uno, motor driver, sensors, battery, and other electronic components are mounted securely on the chassis to ensure proper weight distribution and system reliability.

The front section of the robot contains the ultrasonic and IR sensors, which are responsible for detecting the healthcare worker's position and movement. The mobility system consists of DC motors connected to the wheels, allowing the robot to move forward, turn, and stop as required. A waste bin equipped with a servo-controlled lid is integrated into the robot structure to support touchless medical waste disposal. The overall design is simple, cost-effective, and easy to assemble, making it suitable for prototype development and future improvements.

3.5 Design Alternatives and Rationale

Several design approaches were considered during the development of the Medical Assistive Robot. One possible approach was to use a camera-based human-following system. Although camera-based systems can provide accurate tracking, they require image processing, higher computational power, and increased implementation cost.

Another alternative was the use of RFID or Bluetooth-based tracking methods. These systems require the user to carry additional devices, which may reduce convenience and increase system complexity.

For the waste disposal mechanism, a manual lid system was also considered. However, it would not provide the touchless operation required for maintaining hygiene in healthcare environments.

Therefore, the proposed design uses ultrasonic sensors, IR sensors, DC motors, and a servo motor controlled by an Arduino Uno. This approach was selected because it is cost-effective, easy to implement, requires less processing power, and provides reliable performance for both human-following and touchless waste disposal functions. The selected design successfully balances functionality, simplicity, and affordability.

3.6 Summary

This chapter presented the system analysis and design of the Medical Assistive Robot. The requirements of the system were identified and analyzed to ensure the successful implementation of both human-following and touchless waste disposal functionalities. The overall system were discussed to illustrate how different components interact with each other. In addition, the robot structure and design rationale were presented to justify the selected design approach. The analysis and design process provides a clear foundation for the implementation and operation of the proposed Medical Assistive Robot.

Chapter 4

Methodology and Implementation

4.1 Methodological Framework

The Medical Assistive Robot was developed using a step-by-step embedded system approach. At first, the problem of medical waste handling and hygiene risk in healthcare environments was identified. Healthcare workers often need to move between patients and waste disposal points, which can increase workload and the risk of contamination. Therefore, the project focused on developing a robot that can follow healthcare workers and provide touchless medical waste disposal. After identifying the problem, the required hardware components were selected based on cost, availability, compatibility, and suitability for prototype development. The major components included a microcontroller, ultrasonic sensor, IR sensors, DC motors, L298N motor driver, servo motor, battery, chassis, wheels, dustbin body, and lid mechanism. The system was then divided into smaller modules such as sensing, movement, control, lid operation, power supply, and mechanical structure. Each module was developed and tested separately before final integration. The human-following module was designed using ultrasonic and IR sensors to detect the user's distance and direction. The automatic waste disposal module was designed using an ultrasonic sensor and servo motor to open and close the dustbin lid automatically.

The methodology followed in this project is given below:

1. Identify the medical waste handling problem and define the project objectives.

2. Analyze the functional and non-functional requirements of the Medical Assistive Robot.
3. Select the required hardware components, including microcontroller, ultrasonic sensor, IR sensors, DC motors, motor driver, servo motor, battery, chassis, dustbin body, and connecting materials.
4. Design the robot structure, human-following mechanism, circuit connection, and automatic lid control system.
5. Develop the control logic for sensor reading, human following, motor movement, safe distance control, and automatic lid operation.
6. Test individual modules such as ultrasonic sensor, IR sensors, DC motors, motor driver, servo motor, and power supply.
7. Integrate all modules into the final Medical Assistive Robot prototype.
8. Test the robot in a controlled indoor environment to evaluate human-following movement and touchless waste disposal performance.

4.2 Tools, Libraries, and Technologies Used

The tools, libraries, and technologies used in this project were selected based on simplicity, low cost, availability, and compatibility with Arduino-based robotic system development.

Table 4.1: Tools, Libraries, and Technologies Used

Category	Tool / Technology	Justification
Programming Language	Arduino C/C++ / Embedded C	Used to write the control logic for sensors, motors, and servo-based lid operation.
Development Environment	Arduino IDE	Used for writing, compiling, uploading, and testing the microcontroller program.
Library	Servo.h	Used to control the servo motor for opening and closing the dustbin lid.
Hardware Platform	Arduino UNO / Arduino Nano Microcontroller	Works as the main controller and processes input from sensors to control motors and servo motor.
Distance Sensor	Ultrasonic Sensor	Measures the distance between the robot and the healthcare worker and also detects hand or waste near the bin.
Direction Sensor	IR Sensors	Detect left and right direction to support human-following movement.
Actuators	DC Motors and Servo Motor	DC motors are used for robot movement, and the servo motor is used for lid opening and closing.
Motor Control	L298N Motor Driver	Controls the direction and movement of the DC motors according to microcontroller signals.
Mechanical Parts	Chassis, Wheels, Dustbin Body, Lid	Used to build the physical structure of the robot and support movement and waste disposal.
Power Supply	Battery Pack	Provides power to the microcontroller, sensors, motors, and servo motor.
Circuit Components	Breadboard, Jumper Wires, Connectors	Used to connect electronic components and complete the circuit.

4.3 Flowchart and Algorithm Development

The algorithm of the Medical Assistive Robot follows a continuous sensing and response process. First, the robot initializes all components. Then it reads data from the ultrasonic and IR sensors to detect the position and distance of the healthcare worker. If the user is at a safe distance, the robot moves forward. If the user is too close or not detected, the robot stops. If the left or right IR sensor detects the user's direction, the robot turns accordingly.

For waste disposal, the robot continuously checks whether a hand or waste item is near the bin. When an object is detected near the disposal area, the servo motor opens the lid automatically. After a short delay, the lid closes again. This process repeats continuously while the robot is powered on.

4.3.1 Flow Chart

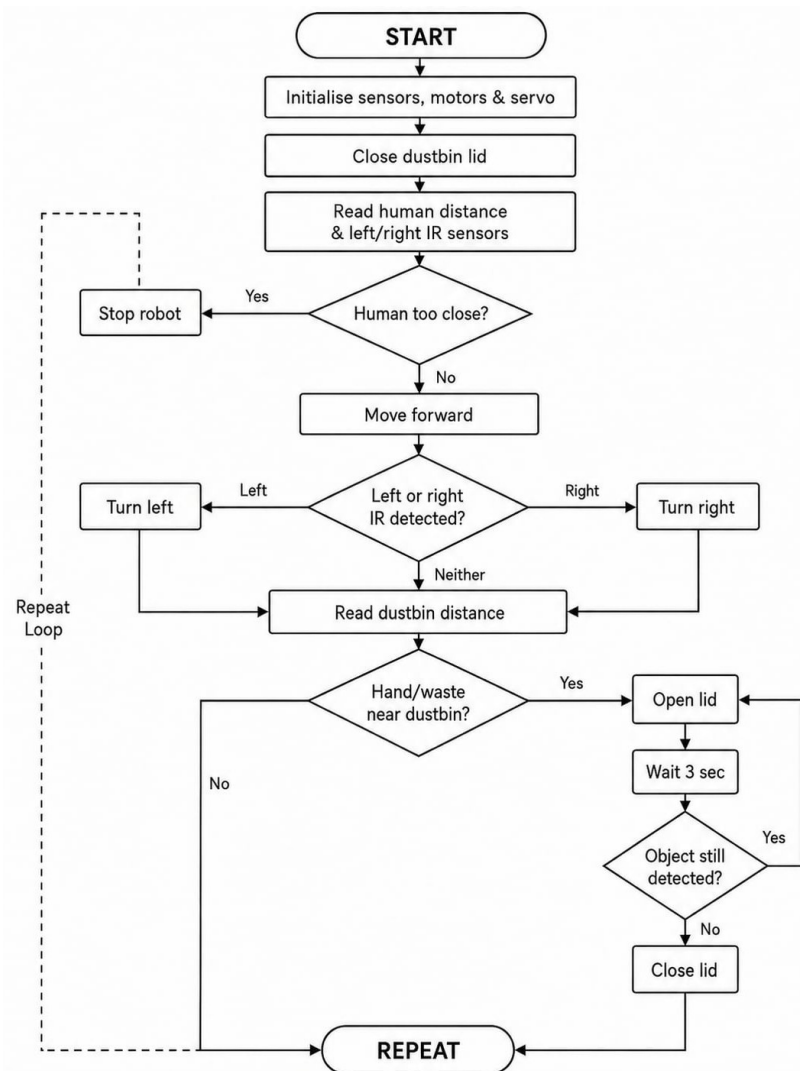


Figure 4.1: Flowchart of the Medical Assistive Robot

4.3.2 Algorithm Development

1. Start.
2. Initialize the microcontroller, ultrasonic sensor, IR sensors, motor driver, DC motors, and servo motor.
3. Set the dustbin lid to closed position.
4. While the robot is powered ON, continuously read sensor values.

5. Read the distance between the robot and the healthcare worker using the ultrasonic sensor.
6. Read left and right IR sensor values for direction detection.
7. If the healthcare worker is too close, stop the robot.
8. Else if the healthcare worker is detected within the following range, move the robot forward.
9. If the left IR sensor detects the user, turn left.
10. Else if the right IR sensor detects the user, turn right.
11. Else continue moving forward or remain stopped based on distance condition.
12. Read the waste detection sensor near the dustbin.
13. If hand or waste is detected near the dustbin, activate the servo motor.
14. Open the dustbin lid automatically.
15. Wait for a short delay.
16. Close the dustbin lid automatically.
17. Repeat the process continuously.
18. End.

4.4 Code

```
{Automatic Dustbin code--}  
  
#include <Servo.h>  
  
  
Servo servo;  
  
  
  
int trigPin = 9;  
int echoPin = 8;
```

```

long duration;
int distance;

bool lidOpen = false;
unsigned long lastSeenTime = 0;
unsigned long waitTime = 3000;

void setup() {
    servo.attach(7);
    servo.write(0);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    Serial.begin(9600);
}

void loop() {
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2;

    if (distance <= 6) {
        servo.write(90);
        lidOpen = true;
        lastSeenTime = millis();
    }
}

```

```

    if (lidOpen && distance > 6) {
        if (millis() - lastSeenTime >= waitTime) {
            servo.write(0);
            lidOpen = false;
        }
    }

    delay(100);
}

```

```

{Human following car--}

```

```

const byte ENA = 9;
const byte IN1 = 5;
const byte IN2 = 6;

```

```

const byte IN3 = 7;
const byte IN4 = 8;
const byte ENB = 10;

```

```

const byte IR_LEFT = 11;
const byte IR_RIGHT = 2;

```

```

const byte TRIG_PIN = 4;
const byte ECHO_PIN = 3;

```

```

int speed = 150;
int minDist = 10;
int maxDist = 40;

```

```

void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(ENB, OUTPUT);

  pinMode(IR_LEFT, INPUT);
  pinMode(IR_RIGHT, INPUT);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  Serial.begin(9600);
}

void loop() {
  int distance = getDistance();

  int leftIR = digitalRead(IR_LEFT);
  int rightIR = digitalRead(IR_RIGHT);

  Serial.print("Distance: ");
  Serial.println(distance);

  if (distance > maxDist) {
    stopCar();
  }
}

```

```

else if (distance > minDist && distance <= maxDist) {

    forward();
}
else {
    stopCar();
}

if (leftIR == LOW && rightIR == HIGH) {
    turnLeft();
}
else if (leftIR == HIGH && rightIR == LOW) {
    turnRight();
}

delay(100);
}

```

```

int getDistance() {
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);

    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    long duration = pulseIn(ECHO_PIN, HIGH);
    int distance = duration * 0.034 / 2;

    return distance;
}

```

```

}

void forward() {
    analogWrite(ENA, speed);
    analogWrite(ENB, speed);

    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);

    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
}

void stopCar() {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, LOW);
}

void turnLeft() {
    analogWrite(ENA, speed / 2);
    analogWrite(ENB, speed);

    digitalWrite(IN1, LOW);
    digitalWrite(IN2, HIGH);

    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
}

void turnRight() {

```

```

analogWrite(ENA, speed);
analogWrite(ENB, speed / 2);

digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);

digitalWrite(IN3, LOW);
digitalWrite(IN4, HIGH);
}

```

4.4.1 Main Modules

The Medical Assistive Robot is divided into several modules so that each part of the system can perform a specific task. These modules work together to follow healthcare workers, maintain safe distance, detect waste, and operate the dustbin lid automatically.

- **Human Detection and Distance Measurement Module:** The human detection and distance measurement module is responsible for detecting the healthcare worker and measuring the distance between the user and the robot. An ultrasonic sensor is used to measure distance. Based on the measured distance, the robot decides whether to move forward, stop, or maintain a safe following distance.
- **Direction Detection Module:** The direction detection module uses IR sensors to identify whether the healthcare worker is positioned toward the left or right side of the robot. If the left IR sensor detects the user, the robot turns left. If the right IR sensor detects the user, the robot turns right. This helps the robot follow the healthcare worker more effectively.
- **Control Module:** The microcontroller works as the main control module of the system. It receives input from the ultrasonic and IR sensors, processes the

data, and sends output signals to the motor driver and servo motor. The control module coordinates human-following movement and automatic lid operation.

- **Movement Module:** The movement module consists of DC motors, wheels, chassis, and the L298N motor driver. The motor driver receives control signals from the microcontroller and controls the direction of the motors. This module allows the robot to move forward, turn left, turn right, and stop when needed.
- **Power Module:** The power module provides the electrical energy required for the operation of the robot. It consists of rechargeable batteries that supply power to the microcontroller, sensors, motors, and other electronic components. A stable power source is essential for maintaining reliable and continuous system performance.
- **Automatic Lid Control Module:** The automatic lid control module is used for touchless medical waste disposal. A servo motor is connected to the dustbin lid. When a hand or waste item is detected near the bin, the microcontroller activates the servo motor to open the lid. After a short delay, the lid closes automatically.
- **Waste Detection Module:** The waste detection module detects whether a hand or waste item is placed near the disposal area. This module helps the system open the dustbin lid without physical contact. It improves hygiene and reduces the risk of cross-contamination in healthcare environments.
- **Robot Chassis and Dustbin Structure Module:** The chassis works as the physical base of the robot. It holds the motors, wheels, sensors, microcontroller, motor driver, battery, and dustbin structure. The dustbin body and lid are mounted on the chassis so that the robot can carry waste and support touchless disposal. A stable structure helps the robot move smoothly and carry the required components.

4.5 Summary

This chapter presents the implementation and performance evaluation of the proposed Medical Assistive Robot. The hardware and software components were integrated to achieve human-following and touchless waste disposal functionalities. The robot was tested under different scenarios to evaluate its ability to follow a healthcare worker, detect waste, and automatically operate the waste bin lid. The results demonstrate that the system can perform its intended tasks effectively while maintaining hygiene and reducing direct contact with medical waste. Overall, the implementation confirms the feasibility of the proposed design and highlights its potential application in healthcare environments.

Chapter 5

Results and Evaluation

5.1 Working Principle

At first, the microcontroller initializes all sensors, motors, and the servo motor. The ultrasonic sensor continuously measures the distance between the robot and the healthcare worker. If the user is within the following range, the robot moves forward. If the user is too close, the robot stops to maintain a safe distance. The IR sensors help detect whether the user is on the left or right side, and the robot turns accordingly.

For waste disposal, another sensor reading is used to detect a hand or waste near the bin. When the object is detected within a short range, the servo motor opens the lid automatically. After a short delay, the lid closes again. This makes the disposal process touchless and more hygienic.

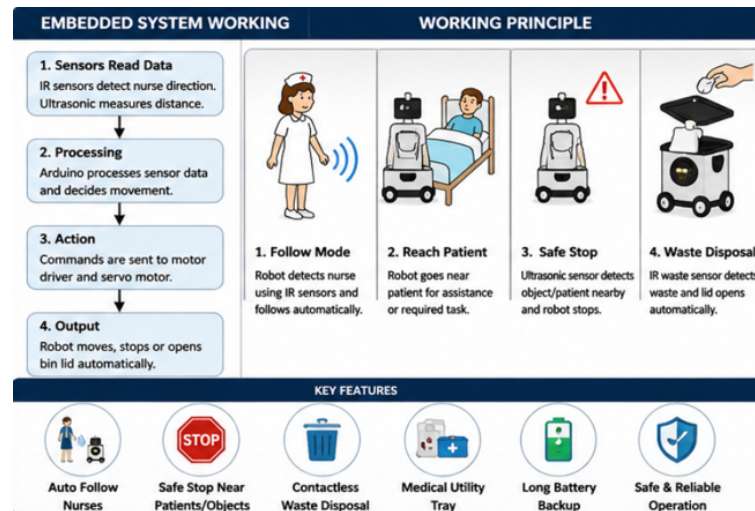


Figure 5.1: Working principle of the Medical Assistive Robot

5.1.1 Data Flow Diagram

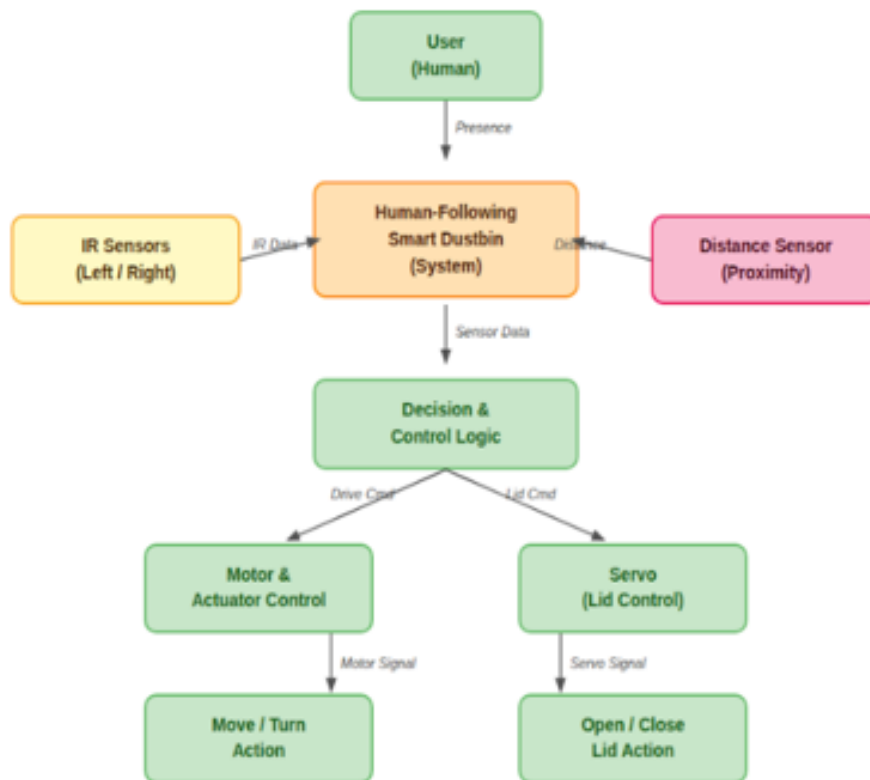


Figure 5.2: Data Flow Diagram of the Medical Assistive Robot

5.2 Planning and Testing

5.2.1 Module Classification and Planning

Objective: Divide the full system into smaller modules for easier development and testing.

- Control Module: Microcontroller for processing inputs and controlling outputs.
- Sensing Module: Ultrasonic and IR sensors for distance and direction detection.
- Actuation Module: DC motors for movement and servo motor for lid operation.
- Power Module: Battery and power connections for stable operation.
- Mechanical Module: Chassis, wheels, dustbin structure, and lid mechanism.

5.2.2 Sensor and Actuator Testing

Objective: Test each component before final integration.

- Ultrasonic Sensor: Tested by measuring different distances and comparing the readings.
- IR Sensors: Tested for left and right detection.
- DC Motors: Tested for forward movement, stopping, and turning.
- Servo Motor: Tested for opening and closing the lid properly.
- Power Supply: Checked to ensure stable power to all components.

5.3 Assembly and Integration

The motors were mounted on the chassis, and the wheels were attached for movement. The ultrasonic sensor was placed at the front side of the robot to measure the user's

distance. IR sensors were placed to detect direction. The servo motor was connected to the dustbin lid so that it could open and close automatically. After completing the hardware setup, all components were connected to the microcontroller and tested together.

5.4 Prototype

The prototype of the Medical Assistive Robot consists of a mobile robotic base with a dustbin structure mounted on it. The robot uses sensors to follow a healthcare worker and a servo-controlled lid for touchless waste disposal. The prototype was designed to demonstrate basic autonomous movement, human-following behavior, and automatic lid operation.

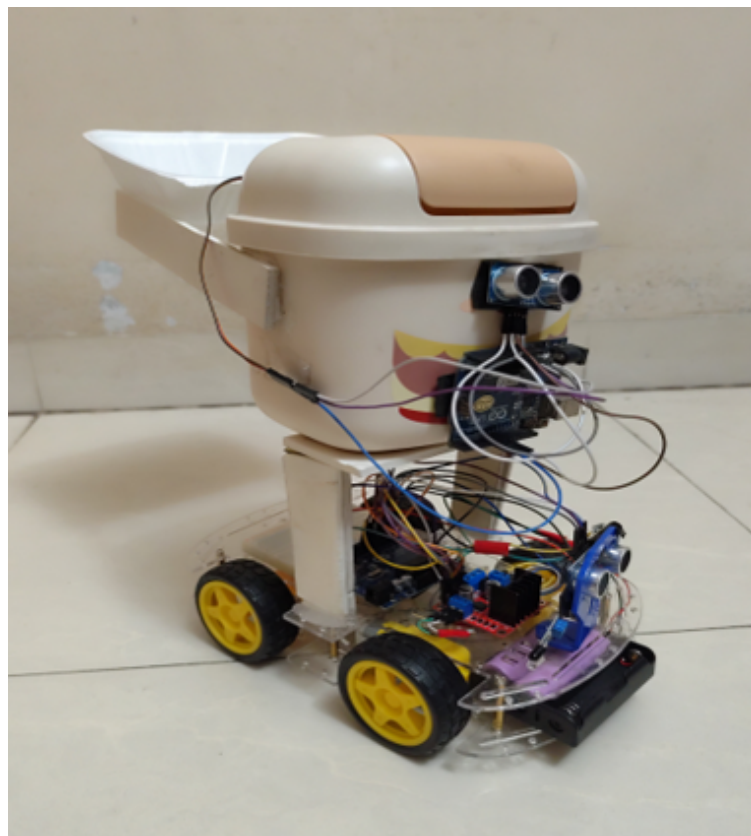


Figure 5.3: Prototype of the Medical Assistive Robot

5.5 Final Design

The final design successfully combines human-following movement and automatic medical waste disposal. The robot can detect the user's distance and direction using ultrasonic and IR sensors. It can move forward, stop, and turn according to the sensor readings. The lid opens automatically when a hand or waste is detected near the bin and closes after a short delay.

5.5.1 Additional Features and Purposeful Engineering Decisions

Additional Features

- **Human-Following Movement:** The robot follows healthcare workers automatically.
- **Automatic Lid Mechanism:** The lid opens without physical contact.
- **Safe Distance Control:** The robot stops when the user is too close.
- **Direction Detection:** IR sensors help the robot turn left or right.
- **Simple Hardware Design:** The system uses low-cost and easily available components.
- **Hygienic Disposal:** Touchless operation helps reduce contamination risk.

Purposeful Engineering Decisions

- **Microcontroller-Based Control:** Chosen for simple and reliable sensor-based operation.
- **Ultrasonic Sensor Use:** Selected for distance measurement between the robot and user.
- **IR Sensor Use:** Selected for direction detection and movement control.

- Servo Motor for Lid: Used because it is simple and suitable for opening and closing the lid.
- Modular Design: The system was divided into sensing, control, actuation, and power modules for easier debugging.
- Low-Cost Design: Components were selected to keep the project practical for a lab-based prototype.

5.6 Performance Evaluation

The system was evaluated based on functional correctness, response behavior, sensor reliability, movement control, and lid operation. The main testing areas were human detection, distance measurement, direction detection, motor movement, safe stopping, waste detection, automatic lid opening, automatic lid closing, and final system integration.

5.7 Result Analysis

The test results show that the ultrasonic sensor successfully measured the distance between the robot and the user. The IR sensors helped detect the direction of the user and supported left and right turning. The robot was able to move forward, stop at a safe distance, and respond to sensor input. The automatic lid mechanism also worked properly during testing. When a hand or waste item was detected near the dustbin, the servo motor opened the lid and closed it after a short delay. The final integrated system successfully demonstrated the two main objectives of the project: human-following movement and touchless medical waste disposal. However, advanced obstacle avoidance and real hospital-level navigation were not fully implemented in the current prototype.

Table 5.1: Functional Test Result

Test Case	Result
Ultrasonic Distance Measurement	Successful
IR Sensor Left Detection	Successful
IR Sensor Right Detection	Successful
Human-Following Movement	Successful
Robot Forward Movement	Successful
Left and Right Turning	Successful
Safe Distance Stop	Successful
Hand/Waste Detection Near Bin	Successful
Servo Motor Lid Opening	Successful
Servo Motor Lid Closing	Successful
Motor Driver Operation	Successful
Power Supply Operation	Successful in prototype testing
Final Integrated System	Successful in controlled indoor testing
Advanced Obstacle Avoidance	Limited / Future Improvement Needed

1. Follow Performance: The robot was able to follow the user based on ultrasonic and IR sensor readings.
2. Movement Control: The robot moved forward, stopped, and turned according to sensor inputs.
3. Lid Operation: The lid opened automatically when a hand or waste was detected near the bin.
4. Hygiene Improvement: Touchless lid operation reduced the need for direct contact with the bin.
5. System Integration: Sensors, motors, and servo motor worked together successfully under microcontroller control.

5.8 Summary

The Medical Assistive Robot shows how a simple robotic system can assist healthcare workers in medical waste handling. The combination of human-following movement and automatic lid operation makes the system useful for hygienic environments such as hospitals and clinics. The robot reduces manual effort because healthcare workers do not need to carry or move the waste bin frequently.

However, the system has some limitations. Sensor readings may be affected by obstacles, surface conditions, or surrounding objects. The following accuracy depends on proper sensor placement and calibration. Battery backup, motor strength, and mechanical stability are also important for better performance. In the future, the robot can be improved by adding better obstacle avoidance, IoT monitoring, stronger chassis design, and more accurate person-following technology.

Chapter 6

Project Planning and Budget

6.1 Project Timeline

Project planning is an important part of engineering design because it helps ensure that all development activities are completed systematically within the available time. The development of the Medical Assistive Robot was carried out through several phases including literature review, requirement analysis, hardware selection, system design, software development, hardware integration, testing, and final documentation. The project began with a detailed study of healthcare robotics, human-following systems, and smart waste management solutions. After identifying the project requirements, suitable sensors, actuators, and control components were selected. Hardware assembly and software implementation were performed simultaneously, followed by repeated testing and debugging to improve system performance. Finally, the complete system was evaluated and documented.

Table 6.1: Project Timeline

Phase	Activities	Duration
Phase 1	Literature Review and Requirement Analysis	Week 1,2
Phase 2	System Design and Component Selection	Week 3
Phase 3	Hardware Assembly	Week 4
Phase 4	Software Development	Week 5
Phase 5	Sensor Testing and Calibration	Week 6
Phase 6	System Integration	Week 7
Phase 7	Performance Evaluation	Week 8
Phase 8	Report Preparation and Final Submission	Week 9

6.1.1 Gantt Chart

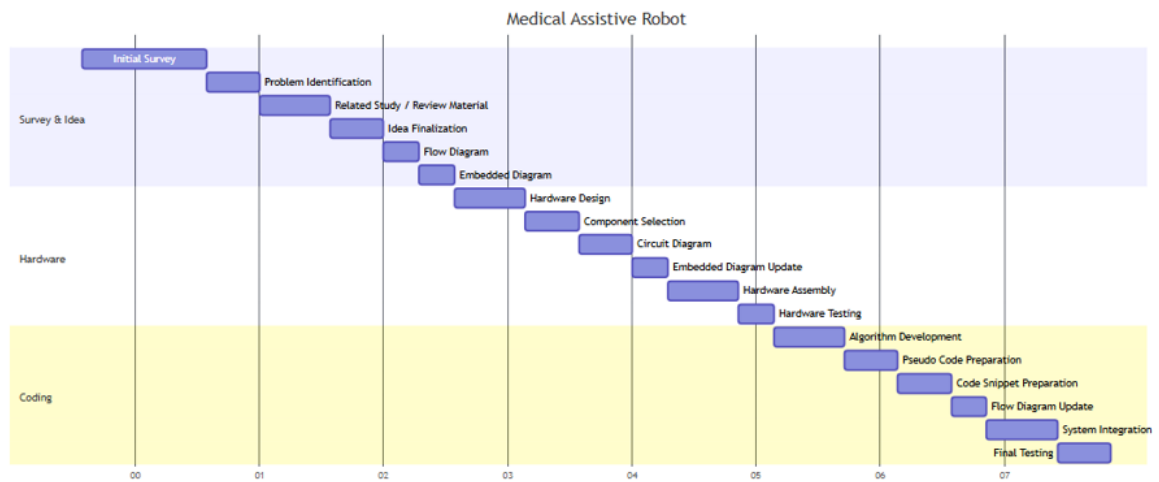


Figure 6.1: GanttChart of the Medical Assistive Robot

6.2 Budget Estimation

Budget estimation was performed to evaluate the economic feasibility of the Medical Assistive Robot. The project was intentionally designed using low-cost and eas-

ily available electronic components so that it could be replicated by students and researchers with limited resources. The major cost elements include the microcontroller, sensors, motors, motor driver, servo motor, battery, chassis, and supporting accessories. Since the robot is intended as a prototype for healthcare applications, affordability and simplicity were considered during component selection.

Table 6.2: Estimated Project Budget

Component	Quantity	Unit Cost (BDT)	Total Cost (BDT)
Arduino Uno	2	900	1800
Ultrasonic Sensor	1	180	180
IR Sensor	2	60	120
Servo Motor	1	300	300
L298N Motor Driver	1	250	250
DC Motors	2	300	600
Chassis and Wheels	1 Set	800	800
Battery Pack	1	500	500
Connecting Wires and Accessories	–	300	300
Miscellaneous Expenses	–	500	500
Total Estimated Cost			5350 BDT

The estimated budget demonstrates that the Medical Assistive Robot can be developed at a relatively low cost while providing practical healthcare assistance functionalities. This affordability increases the potential for educational use and future development in healthcare robotics research.

6.3 Summary

This chapter presented the project planning and budget estimation of the Medical Assistive Robot. A structured development timeline was followed to ensure systematic implementation of the project. In addition, a cost analysis was conducted to evaluate the financial feasibility of the prototype. The results indicate that the proposed system can be implemented using affordable components while maintaining the required functionality for healthcare assistance and touchless waste management.

Chapter 7

Standards and Design Constraints

7.1 Compliance with Standards and Professional Practice

The design and development of the Medical Assistive Robot were carried out following standard engineering practices related to embedded systems, robotics, healthcare safety, and software development. The project emphasizes reliability, safety, maintainability, and ethical responsibility while addressing a real-world healthcare problem.

7.1.1 Software Standards

The software system was developed using Embedded C/C++ programming language within the Arduino IDE environment. Structured programming principles were followed to improve code readability, modularity, maintainability, and future scalability. The software architecture was divided into sensing, decision-making, movement control, and lid-control modules to ensure efficient debugging and future expansion.

7.1.2 Hardware Standards

The hardware design follows common embedded system design principles. Standard voltage levels were maintained for sensors, actuators, and microcontroller components. Proper electrical connections, power distribution, and component integration were

ensured to minimize operational failures. The selected hardware components are commercially available and widely used in educational and industrial prototyping applications.

7.1.3 Safety and Ethical Considerations

Since the robot is intended for healthcare environments, safety and hygiene were considered important design priorities. The touchless waste disposal mechanism minimizes direct contact with medical waste containers, thereby reducing contamination risks. The robot does not collect, store, or process personal patient information, ensuring privacy protection and ethical compliance.

7.1.4 Environmental and Sustainability Considerations

The system utilizes low-power electronic components that consume minimal energy during operation. The use of rechargeable batteries can further reduce environmental impact. In addition, proper medical waste handling supported by the robot contributes to a cleaner and safer healthcare environment.

7.1.5 Professional Engineering Practice

The project follows the engineering design cycle involving requirement analysis, system design, implementation, testing, evaluation, and documentation. Decisions were made based on technical feasibility, safety requirements, and practical healthcare needs.

7.2 Design Constraints

Several constraints influenced the design and implementation of the Medical Assistive Robot.

7.2.1 Technical Constraints

The tracking performance depends on the operating range and accuracy of the ultrasonic and IR sensors. Sensor noise, environmental conditions, and lighting variations may affect detection accuracy. The processing capability of the microcontroller also limits the implementation of advanced artificial intelligence or computer vision techniques.

7.2.2 Economic Constraints

The project was developed using a limited budget. Therefore, low-cost sensors, actuators, and control hardware were selected instead of more sophisticated industrial-grade equipment.

7.2.3 Operational Constraints

The robot is designed primarily for indoor healthcare environments. Its performance may decrease in crowded spaces, uneven surfaces, or highly dynamic environments. Battery capacity also limits the continuous operating duration of the system.

7.2.4 Health and Safety Constraints

The robot must operate safely around healthcare workers and patients. Excessive speed, unstable movement, or unexpected mechanical actions could create safety concerns. Therefore, controlled movement and stable operation were prioritized during development.

7.3 Complex Engineering Problem (CEP) Coverage

The development of the Medical Assistive Robot involves solving a complex engineering problem because it requires the integration of sensing, embedded control, robotics,

and healthcare-oriented automation within a single system. The project combines multiple hardware and software components to achieve reliable human-following behavior and touchless medical waste disposal under practical operating conditions. The robot must continuously process data from ultrasonic and IR sensors, determine the user's position and movement direction, maintain a safe following distance, and simultaneously control the automatic lid mechanism. These requirements involve multiple interacting subsystems and engineering trade-offs related to cost, performance, safety, reliability, and usability.

Table 7.1: Mapping with Complex Engineering Problems (Ps)

CEP At- tribute	Description	Application in Medical Assistive Robot
P1	Depth of Knowledge Required	Integration of robotics, embedded systems, sensors, and healthcare automation concepts.
P2	Range of Conflicting Requirements	Balancing cost, accuracy, power consumption, and reliability.
P3	Analysis of Complex Problems	Processing sensor data and decision-making for human-following behavior.
P4	Familiarity of Issues	Healthcare-assistive robotics requires adaptation of existing technologies.
P5	Extent of Applicable Codes	Consideration of healthcare safety and engineering standards.
P6	Stakeholder Involvement	Healthcare workers, patients, and hospital management are affected.
P7	Interdependence	Interaction among sensing, control, movement, and disposal subsystems.

The mapping demonstrates that the Medical Assistive Robot satisfies the characteris-

tics of a Complex Engineering Problem by requiring multidisciplinary knowledge and systematic engineering analysis.

7.4 Complex Engineering Activities (CEA) Coverage

The Medical Assistive Robot also involves complex engineering activities because the project required investigation, design, implementation, testing, evaluation, and optimization of an integrated robotic system. The development process included hardware selection, circuit integration, software development, sensor calibration, system testing, and performance evaluation.

Table 7.2: Mapping with Complex Engineering Activities (As)

CEA At- tribute	Description	Application in Medical Assistive Robot
A1	Range of Resources	Use of sensors, actuators, microcontroller, software tools, and healthcare knowledge.
A2	Level of Interaction	Interaction between sensing, movement, and waste disposal modules.
A3	Innovation Requirement	Integration of human-following and touchless waste disposal into one platform.
A4	Consequences to Society	Improved hygiene and reduced workload in healthcare environments.
A5	Familiarity	Adaptation of existing robotic technologies for healthcare applications.

The project demonstrates complex engineering activities through system integration, practical implementation, testing, and evaluation of a healthcare-oriented robotic solution.

7.5 Knowledge Profile Mapping

The Medical Assistive Robot incorporates knowledge from several engineering disciplines. The project requires the application of mathematics, programming, electronics, sensor technology, robotics, embedded systems, and system integration principles.

Table 7.3: Knowledge Profile Mapping

Knowledge Profile	Application in Medical Assistive Robot
WK1 – Natural Sciences	Sensor operation and signal measurement.
WK2 – Mathematics	Distance calculation and decision logic.
WK3 – Engineering Fundamentals	Electronic circuits and motor control.
WK4 – Specialist Knowledge	Robotics and embedded system design.
WK5 – Engineering Design	Development of human-following and waste disposal system.
WK6 – Engineering Practice	Hardware assembly, testing, and troubleshooting.
WK7 – Society and Environment	Healthcare safety and hygiene improvement.
WK8 – Ethics	Safe and responsible use in healthcare environments.

7.6 Summary

This chapter showed how the Medical Assistive Robot covers the required knowledge profiles, complex engineering problems, and complex engineering activities. The project combines microcontroller-based embedded control, human-following navigation, ultrasonic and IR sensor integration, motor control, automatic lid operation,

hardware assembly, software development, testing, and documentation. It also considers practical issues such as healthcare hygiene, safety, cost-effectiveness, power consumption, component selection, system reliability, and ease of operation. The robot integrates multiple subsystems to provide both human-following assistance and touchless medical waste disposal in healthcare environments. Overall, the project demonstrates both theoretical engineering knowledge and practical engineering application through the design, development, implementation, and evaluation of a functional Medical Assistive Robot prototype.

Table 7.4: Overall Mapping of Knowledge Profile, CEP and CEA

Category	Covered At- tributes	Project Evidence
Knowledge Profile	WK1– WK8	Applied engineering fundamentals, sensor operation, embedded programming, motor control, robotics concepts, system integration, healthcare safety considerations, testing, documentation, and literature review.
Complex Engineering Problems	P1–P7	Addressed human-following navigation, sensor-based decision making, healthcare waste management, hardware-software integration, cost limitations, safety requirements, stakeholder needs, and interaction among multiple system modules.
Complex Engineering Activities	A1–A5	Included component selection, circuit design, hardware assembly, sensor integration, automatic lid control, software implementation, testing, evaluation, resource management, and safety-focused system development.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

The Medical Assistive Robot project successfully demonstrates a smart and hygienic medical waste management system. It combines ultrasonic and IR sensor-based human-following movement with an automatic lid-opening mechanism for touchless disposal. The robot can support healthcare workers by reducing manual effort, improving cleanliness, and minimizing the risk of cross-contamination.

Overall, this project is a practical example of using robotics, sensors, and microcontroller-based automation to solve a real-life healthcare problem. Future improvements may include advanced obstacle avoidance, longer battery life, IoT-based monitoring, and improved mechanical design.

8.2 Future Work

Table 8.1: Future Work of the Medical Assistive Robot

Future Work	Description
• Voice Control System	The robot can follow voice commands from users or hospital staff.
• Mobile App Monitoring	Users can monitor and control the robot using a smartphone app.
• Camera Monitoring System	A camera can help monitor patients and surroundings for better safety.
• GPS / Indoor Navigation	The robot can move automatically to specific rooms or hospital locations.
• IoT Connectivity	Robot data and status can be monitored through the internet in real time.
• Emergency Alert System	The robot can send alerts during emergency situations in hospitals.
• Automatic Waste Segregation	The dustbin can automatically separate medical and normal waste.

References

- [1] J Sattar Md Jahidul Islam, J Hong and Junaed. Person following by autonomous robots: A categorical overview. 2018.
- [2] Lei Shi Yong Liu Jaime Valls Miro Mengmeng Wang, Daobilige Su. Real-time 3d human tracking for mobile robots with multisensors. 2017.
- [3] R.F.A. Bonifacio. Applications of robotic systems and emerging technologies in the collection, classification, and treatment of medical waste: A systematic review. 2025.
- [4] Zhiqiang Zhou; Yong Fu; Wei Wu. Human-following task without a prior map. 2024.

Appendices

A Surveys

A.1 Survey 1

Responses cannot be edited

Survey Form 1

This data will only be used for research purpose for the academic class. The data will be destroyed after the research is finished. No form of personal information sharing will be done.(এই তথ্য শুধুমাত্র ক্লাসের গবেষণার উদ্দেশ্যে ব্যবহার করা হবে। গবেষণা শেষ হওয়ার পরে তথ্যটি সরিয়ে ফেলা হবে। কোনও ধরনের ব্যক্তিগত তথ্য ফাঁস করা হবে না।)

* Indicates required question

Your Name (Only if you are okay to share)

Mim Akter

What is your job role? (চাকরির পদ) *

nurse

Are you involved in purchasing/decision making? (আপনি কি ক্রয়/সিদ্ধান্ত গ্রহণের সাথে জড়িত?) *

☒ Yes

☐ No

Have you understood the product concept? (আপনি কি পণ্যের ধারণাটি বুঝতে পেরেছেন?) *

12345

Did not understand

☐

☐

☐

☐

☒

Fully understand

Do you think our product will be helpful for your workplace? (আপনি কি মনে করেন আমাদের পণ্য আপনার কর্মক্ষেত্রের জন্য সহায়ক হবে?) *

1 2 3 4 5

Not Helpful ☐ ☐ ☐ ☐ ☒ Very Helpful

Additional comments (অতিরিক্ত মন্তব্য)

It would be better if work perfectly as it claim to be.

Submitted 4/14/26, 11:08 PM

A.2 Survey 2

Responses cannot be edited

Survey Form 2

This data will only be used for research purpose for the academic class. The data will be destroyed after the research is finished. No form of personal information sharing will be done.(

এই তথ্য শুধুমাত্র ক্লাসের গবেষণার উদ্দেশ্যে ব্যবহার করা হবে। গবেষণা শেষ হওয়ার পরে তথ্যটি সরিয়ে ফেলা হবে। কোনও ধরনের ব্যক্তিগত তথ্য ফাঁস করা হবে না।)

* Indicates required question

Your Name (Only if you are okay to share)

Mim Akter

What is your job role? (চাকরির পদ) *

Nurse

What is your job role? (চাকরির পদ) *

Nurse

Are you involved in purchasing/decision making? (আপনি কি ক্রয়/সিদ্ধান্ত গ্রহণের সাথে জড়িত?) *

☒ Yes

☐ No

Have you understood the product concept? (আপনি কি পণ্যের ধারণাটি বুঝতে পেরেছেন?) *

1 2 3 4 5
Did not understand ☐ ☐ ☐ ☐ ☒ Fully understand

Do you think our product will be helpful for your workplace? (আপনি কি মনে করেন আমাদের পণ্য আপনার কর্মক্ষেত্রের জন্য সহায়ক হবে?) *

1 2 3 4 5
Not Helpful ☐ ☐ ☐ ☒ Very Helpful

Would you purchase this product? *
(আপনি কি এই পণ্যটি কিনবেন?)

1 2 3 4 5
Not Helpful ☐ ☐ ☐ ☒ Very Helpful

Do you think the product price is acceptable? *
(আপনি কি মনে করেন পণ্যটির মূল্য গ্রহণযোগ্য?)

1 2 3 4 5
Not Helpful ☐ ☐ ☐ ☒ Very Helpful

How much money would you spend for this ind of product? (*
এই ধরনের পণ্যের জন্য আপনি কত টাকা খরচ করবেন?)

5-6 thousand taka

How much money would you spend for this ind of product? (*
এই ধরনের পণ্যের জন্য আপনি কত টাকা খরচ করবেন?)

5-6 thousand taka

Which part of this product needs improvement? (এই পণ্যটির কোন অংশে উন্নতি প্রয়োজন?) *

Structure and response speed

Additional comments (অতিরিক্ত মন্তব্য)

Overall good

Submitted 6/24/26, 11:33 PM



Figure 8.1: Prototype feedback interview with Nurse

B Team Contributions

ID	Name	Contribution
22201176	Meher Afroz	• Developed autonomous movement logic. • Implemented sensor-based decision making (ultrasonic + IR). • Designed algorithm, flowchart and pseudocode. • Developed control workflow and system logic. • Worked on testing and result analysis.
22201184	Hridita Lasker	• Conducted survey and idea development. • Performed literature review and background study. • Designed system architecture and embedded diagram. • Handled report formatting and documentation. • Prepared figures, tables and final editing.
22201194	Jannatul Maowa Meghla	• Designed hardware system and component selection. • Integrated sensors, motors and microcontroller. • Developed circuit diagram and wiring system. • Implemented servo-based automatic lid mechanism. • Built and tested prototype system.

Table 8.2: Team Contributions in Medical Assistive Robot Project

C Ethics and Compliance Statements

This Medical Assistive Robot was developed as an academic prototype and tested only in a controlled environment. The following ethical considerations were maintained:

- The system does not collect or store any personal or patient data.
- The robot is designed only for academic and prototype purposes.
- Proper safety measures were taken during hardware testing.
- The system helps reduce contact with medical waste and improves hygiene.
- All results are reported honestly without manipulation.
- The system is not intended to replace professional medical systems.